

Fig. 3: Simulation flow for vehicle powernet systems

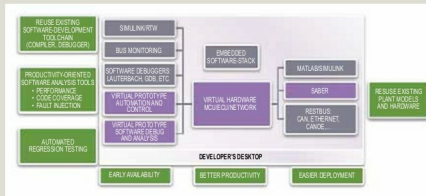


Fig. 4: Virtual prototyping of automotive systems

to evaluate design architecture trade-offs—be it parallel, serial or any other complex topology—along with analysis of power generation and distribution for motor drives and controls, regenerative braking, power assist, etc. Saber also incorporates multi-domain (mechanical, electrical, thermal and magnetics) effects at the system level. Designers can design, test and verify

control strategies, power management, torque/speed coupling and vehicle dynamics while being able to optimise the cost, system performance and reliability with advanced sensitivity, stress and statistical analyses.

Wire harness design and simulation. Wire harnesses form the backbone of the entire electrical system of automotive and aerospace vehicles. Their reliable

implementation represents one of the most expensive and technically challenging aspects of vehicle systems design. Saber harness provides proven design and verification capabilities in conjunction with Saber simulator to create correct-by-design wire harnesses. Designers can also create schematic drawings and connectivity diagrams, export components and wire data, import required geometry information from MCAD tools, simulate electrical functions, create bundles with connector positions and generate data for manufacturing—all within an easy-to-use design tool.

Saber advantages for wire harness design and simulation. Saber lets designers analyse electrical systems before layout and manufacturing, eliminating system failure in production. It also provides an integrated data flow for electrical system design from concept to manufacturing. It minimises data entry and manual checking tasks, and automates data processing steps while maintaining data integrity. The software integrates seamlessly with popular 3D CAD tools like Catia V5, UGS and Pro/E.

View the possibilities

To get the first-hand impression of Synopsys Saber simulator for automotive sector, test-run it from the EFY DVD. Please note that you need at least 1.4 GB of free space on your hard disk for the installation. More in-depth documentation and installation guides are included in the DVD, which you may find useful. ●

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